

Note on a Theorem of Indeterminate Analysis

I read in the summary of the meetings of the Académie des Sciences, inserted in the *Annales de Chimie et de Physique*: “M. Lamé presents a memoir on the impossibility of the equation $x^5 + y^5 = 2^n z^5$, in integers.” This recalls to me a theorem of the same kind on which I came upon more than a year ago, and whose statement follows.

The equation

$$x^3 \pm y^3 = 2^n z^3$$

is impossible in positive integers.

To prove this theorem it will be enough to prove the impossibility of the three equations

$$x^3 \pm y^3 = z^3, \quad x^3 \pm y^3 = 2z^3, \quad x^3 \pm y^3 = 4z^3. \quad (\text{a})$$

For if one wished to suppose $x^3 \pm y^3 = 2^n z^3$, with $n = 0$ or $n > 3$, one need only put $n = 3k + r$, where r is one of the numbers 0, 1, 2; this changes the preceding equation into

$$x^3 \pm y^3 = 2^r (2^k z)^3,$$

which falls under one of the equations (a) assumed impossible.

The impossibility of the first two equations (a) has long since been proved, as may be seen in the second volume of EULER’s *Algebra* or in M. LEGENDRE’s theory of numbers. I therefore have only to deal with the third.

Suppose, if possible, that $x^3 + y^3 = 4z^3$. Since we may suppose that x and y are relatively prime, they will both be odd. Put $x = p + q$ and $\pm y = p - q$; then p and q are integers, relatively prime, and moreover one is even and the other odd. Substitution gives

$$p(p^2 + 3q^2) = 2z^3,$$

which shows that p must be even, for otherwise the first member would be odd; hence $p/2$ is an integer and

$$\frac{p}{2}(p^2 + 3q^2) = z^3.$$

There are now two cases, according as p is or is not divisible by 3.

First case. If p is not divisible by 3, then $p/2$ and $p^2 + 3q^2$ are relatively prime; since their product is a cube, each of these numbers must be a cube. To make $p^2 + 3q^2$ a cube in the most general way, one need only put, as is known,

$$p = m^3 - 9mn^2, \quad q = 3m^2n - 3n^3.$$

From what has been said about p and q , it is easily seen that m must be prime to n , and moreover even and not divisible by 3. It follows that the three factors in

$$\frac{p}{2} = \frac{m}{2}(m + 3n)(m - 3n)$$

are relatively prime. Since their product $p/2$ is to be a cube, each of them must be a cube; consequently

$$\frac{m}{2} = \alpha^3, \quad m + 3n = \beta^3, \quad m - 3n = \gamma^3.$$

This gives $\beta^3 + \gamma^3 = 4\alpha^3$, an equation of the same form as $x^3 + y^3 = 4z^3$, but expressed in much smaller numbers.

Second case. If p is divisible by 3, put $p = 3r$; then r , like p , is even and relatively prime to q . Thus

$$\frac{9r}{2}(q^2 + 3r^2) = z^3.$$

It is easy to see that the two factors of the first member are relatively prime; hence each must be a cube. To make $q^2 + 3r^2$ equal to a cube, put, as in the first case,

$$q = m^3 - 9mn^2, \quad r = 3m^2n - 3n^3.$$

Then n must be prime to m , and even; hence $9r/2$ is an integer and

$$\frac{9r}{2} = 27 \frac{n}{2} (m+n)(m-n).$$

The factors $n/2$, $m+n$, and $m-n$ are relatively prime. Since their product multiplied by the cube 27 is to be a cube, each of them must be a cube. Thus

$$\frac{n}{2} = \alpha^3, \quad m+n = \beta^3, \quad m-n = \gamma^3,$$

and therefore $\beta^3 - \gamma^3 = 4\alpha^3$, an equation similar to the proposed equation $x^3 \pm y^3 = 4z^3$, but expressed in much smaller numbers.

Thus, by the ordinary infinite descent, the proposed equation is not possible.